

Tide-experiment addendum:
SGLD bias on κ_3 is linear in α with slope ≈ 0.22
(non-monotonicity in the main run was MC noise)

Claude (Opus 4.7, 1M context)

2026-05-23 (continuation of the same-day experiment)

Abstract

The main retrospective¹ reported a non-monotonic SGLD bias $\Delta_{\text{SGLD-HMC}} \in \{+0.092, +0.060, +0.277\}$ across $\alpha \in \{0.25, 0.5, 1.0\}$, drawn from a coarse 64-chain $\times$ 20k-draw HMC and 64-chain $\times$ 40k-draw SGLD α -scan as a mechanism probe. At higher precision (64 chains $\times$ 40k draws HMC, 64 chains $\times$ 80k draws SGLD) the non-monotonicity disappears: $\Delta_{\text{SGLD-HMC}}$ scales *monotonically and approximately linearly* with α , slope ≈ 0.22 at $\alpha \geq 0.5$. The original non-monotonicity is identified as MC noise.

For Patterning 2 three-point susceptibilities — which use a third-order cumulant on a model with intrinsic asymmetry α — this gives a practical rule of thumb: *SGLD’s pipeline-default* ($\varepsilon = 5 \times 10^{-3}$) *bias on κ_3 at WBIC temperature is $\approx 0.22\alpha$.*

1 The denser α -scan

Same model ($L(x) = (\lambda/2)x^2 + (\alpha/6)x^3 + (\gamma/24)x^4$ with $\lambda = 1, \gamma = 1$) and same headline cell $t = 100$ as the main retrospective. Six α values, all satisfying the $\alpha^2 < 3\lambda\gamma$ discriminant condition. Chains: 64 $\times$ 40k draws HMC, 64 $\times$ 80k draws SGLD.

α	formal $-\alpha/\lambda^3$	HMC $t^2\hat{\kappa}_3 \pm \text{SE}$	HMC Δ vs formal	SGLD $t^2\hat{\kappa}_3 \pm \text{SE}$	$\Delta_{\text{SGLD-HMC}} \pm \text{SE}$	Δ/α
0.10	−0.10	-0.1016 ± 0.0144	$+0.002$ (0.1 σ)	-0.0703 ± 0.0216	$+0.031 \pm 0.026$ (1.2 σ)	0.31
0.25	−0.25	-0.2303 ± 0.0134	$+0.020$ (1.5 σ)	-0.2148 ± 0.0163	$+0.016 \pm 0.021$ (0.7 σ)	0.06
0.50	−0.50	-0.5003 ± 0.0154	-0.000 (0.0 σ)	-0.3929 ± 0.0191	$+0.107 \pm 0.025$ (4.4 σ)	0.21
0.75	−0.75	-0.7359 ± 0.0166	$+0.014$ (0.8 σ)	-0.5726 ± 0.0198	$+0.163 \pm 0.026$ (6.3 σ)	0.22
1.00	−1.00	-0.9847 ± 0.0137	$+0.015$ (1.1 σ)	-0.7613 ± 0.0176	$+0.223 \pm 0.022$ (10.0 σ)	0.22
1.40	−1.40	-1.4602 ± 0.0157	-0.060 (3.8 σ)	-1.1358 ± 0.0226	$+0.324 \pm 0.028$ (11.7 σ)	0.23

2 The linearity, in plain prose

For $\alpha \geq 0.5$, the ratio $\Delta_{\text{SGLD-HMC}}/\alpha$ is flat at ≈ 0.22 across four cells. For $\alpha \leq 0.25$ the absolute bias is small relative to MC SE (the bias-to-SE ratio is ≤ 1.2), and the apparent Δ/α value (0.31 and 0.06) is dominated by noise — the bias is below the precision floor at those α values.

Conclusion: $\Delta_{\text{SGLD-HMC}}(\alpha; t = 100, \varepsilon = 5 \times 10^{-3}) \approx 0.22\alpha$ at moderate-to-large α .

¹2026-05-23-experiment-kappa3-anharmonic-1d.

The main run's apparent $\Delta_{\text{SGLD-HMC}} = +0.060$ at $\alpha = 0.5$ was 1.5 SE below the trendline ($0.22 \cdot 0.5 = 0.11$); this addendum's denser run gives $+0.107$, right on the trend. Similarly the main run's $+0.277$ at $\alpha = 1.0$ was 1.6 SE above; this run gives $+0.223$. *The "non-monotonic" pattern was MC noise.*

3 Why this matters

The Patterning 2 three-point susceptibility framework uses κ_3 on a model with intrinsic cubic-asymmetry strength α (the cubic susceptibility of the network's loss landscape). At the seabed's SGLD pipeline defaults this addendum predicts:

$$\hat{\kappa}_3^{\text{SGLD}} \approx \kappa_3^{\text{true}} + 0.22\alpha \quad (\text{at } t = 100, \varepsilon = 5 \times 10^{-3}).$$

For Patterning 2 to use κ_3 as a quantitative susceptibility, either (a) the experiment runs in a regime where $0.22\alpha \ll \kappa_3$ (weak intrinsic skew); (b) the SGLD is run at $\varepsilon \leq 5 \times 10^{-4}$ where the main retrospective's ε -scan showed the bias vanishing; (c) HMC is used instead.

4 HMC vs formal at large α

A second observation worth noting: HMC's $t^2 \hat{\kappa}_3(\alpha = 1.4) = -1.460$ vs formal asymptote -1.40 is -0.06 off, a 3.8σ deviation. At $\alpha = 1.0$ HMC is $+0.015$ above formal (1.1σ), also non-trivial. At $\alpha \leq 0.75$ HMC matches within 1σ .

This is consistent with the *finite-t correction* predicted by the next-order Laplace expansion. The main retrospective's pre-check showed the correction $+1.25/t$ at $\alpha = 0.5$; the coefficient scales with α (and possibly higher powers). At $\alpha = 1.4$, the expected next-order shift is roughly $+1.25 \cdot (1.4/0.5)/t = +0.035$ if linear-in- α , putting the finite- t value at -1.365 — but observed is -1.46 . The deviation is -0.10 , suggesting the next-order coefficient grows *super*-linearly in α , or that a higher-order term ($O(\alpha^3/t)$?) becomes visible.

This is a candidate for a formalisation tide: extend Tide 9 in laplace to compute the next-order coefficient $c_1(\alpha, \lambda, \gamma)$ in $t^2 \kappa_3(t) = -\alpha/\lambda^3 + c_1/t + O(1/t^2)$ and check the empirical α -scaling.

5 Data

JSONL output: `local/kappa3-anharmonic-1d/data/alpha_scan_tight.jsonl`. Each row records HMC and SGLD $\hat{\kappa}_3$, SE, and the $\Delta_{\text{SGLD-HMC}}$ at the given α . Compute: 6 cells $\times$ (15.7s HMC + smaller SGLD) $\approx$ 1.5 minutes total.

Acknowledgements

Continuation of the same-day 2026-05-23-experiment-kappa3-anharmonic-1d κ_3 tide-experiment. The non-monotonicity in the main run turned out to be MC noise, illustrating the importance of the cherry-pick guard the tide-experiment skill prescribes (report SE alongside any "best of n cells" claim and re-run dense to confirm).